

Hotel Bar Inventory Forecasting & Par Level Recommendation System

Business Report | AI/ML Intern Assignment

1. Executive Summary

This project develops an inventory forecasting and par-level recommendation system for hotel bars. The solution converts transaction-level bottle balance records into daily consumption time series, evaluates forecasting methods, estimates dynamic inventory requirements, and tests the resulting inventory policy through backtesting.

The dataset contains 6,575 transaction records covering January 1, 2023 to January 1, 2024, across 6 bars, 16 brands, and 5 alcohol categories. Holt-Winters Exponential Smoothing produced the lowest MAE and WAPE among the four evaluated forecasting approaches and was selected for dynamic par-level calculation.

2. Data Preparation & Quality

The data was checked for missing values, duplicates, invalid dates, negative quantities, and inventory conservation. Closing Balance = Opening Balance + Purchase – Consumed was validated for all 6,575 records within a 10 ml tolerance. Daily consumption was aggregated by date, bar, and brand, with missing dates represented as zero-consumption observations.

3. Forecasting Results

Model	MAE (ml)	WAPE (%)
7-Day Rolling Mean	94.44	169.52
7-Day Seasonal Naive	97.53	175.06
Holt-Winters	91.91	164.97
Random Forest	95.02	170.56

Holt-Winters was selected because it achieved the lowest observed MAE (91.91 ml) and WAPE (164.97%) on the chronological test set. The evaluation uses an 80/20 temporal split to preserve forecasting time order.

4. Dynamic Par-Level Recommendation

The policy uses Par Level = Forecasted Demand during Lead Time + Safety Stock. A 2-day lead time and 95% service-level Z value of 1.645 were used. Safety stock was based on forecast error variability, so the recommendation accounts for both expected demand and forecast uncertainty.

5. Inventory Simulation & Key Results

Metric	Result
Bar-Brand combinations simulated	16
Simulation period	74 days
Bar-Brand-day observations	1,184
Simulated stockout days	47
Simulated stockout observation rate	3.97%
Simulated lost volume	5,149.46 ml
Combinations with zero simulated stockouts	7 of 16
Combinations requiring attention	9 of 16

The simulation highlights differences between individual Bar-Brand combinations. Taylor's Bar — Jim Beam recorded 4 stockout days and 1,186.63 ml of simulated lost volume, while Taylor's Bar — Sutter Home recorded 9 stockout days and 997.21 ml. Johnson's Bar — Miller also recorded 9 stockout days. These cases can be prioritized for replenishment and lead-time review.

6. Historical Inventory Depletion Signals

There were 416 historical records with zero closing balance. Of these, 102 also recorded positive consumption, with 26,363.84 ml of associated consumption. These observations are treated as inventory depletion signals, not confirmed lost sales. Historical and simulated rates use different denominators and should not be interpreted as a direct percentage reduction.

7. Business Recommendations

- Use dynamic Bar-Brand-specific par levels instead of fixed inventory thresholds.
- Review high-risk combinations with repeated simulated stockouts and prioritize replenishment review.
- Use ABC classification to focus attention on high-consumption brands.
- Recalculate par levels periodically as demand patterns and forecast errors change.
- Monitor stockouts, lost volume, average inventory, orders, and turnover after deployment.
- Review supplier lead times where stockouts persist despite replenishment.

8. Limitations

The dataset contains inventory transactions rather than confirmed customer-level lost-sales observations. Therefore, zero closing balance is a depletion signal rather than proof of lost demand. Forecast accuracy can also differ by Bar-Brand combination because demand volume and

variability are not uniform.

9. Conclusion

The solution demonstrates an end-to-end workflow from inventory validation and demand aggregation to forecasting, dynamic par-level calculation, and inventory policy simulation. Holt-Winters provided the strongest test-set performance among the evaluated models, and the simulation identifies specific Bar-Brand combinations requiring additional inventory management attention.